

Husam Kassab

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EXPERIENCE

Kenna Technology Solutions | Full Stack Application Developer

September – December 2024

- Implemented **UI/UX** design based on user feedback, enhancing usability and engagement
- Developed and optimized **RESTful APIs** using **NodeJS** and **ExpressJS**, reducing response times by **25%**
- Designed **SQL** database tables to improve query performance and data integrity, increasing system efficiency
- Collaborated with a large team using **Git** and **Bitbucket** to streamline workflows and maintain release schedules
- Engineered responsive front-end components with **React** and **Redux**, reducing initial page load time by **20%** and improving cross-device consistency

RELEVANT PROJECTS

Large Language Model Analyzer

April 2025

- Developed a full-stack AI chat application that enabled sub-second real-time messaging by leveraging **React.js**, **Node.js**, **Express.js**, **MongoDB**, and **Socket.IO**
- Integrated models such as **Claude-3.5**, **GPT-3.5**, **GPT-4o-mini**, **o3-mini**, **o1-mini**, boosting response relevance and flexibility through **Few-Shot**, **Chain-of-Thought**, and **Role-based prompting**
- Built a **PyTorch** cost estimator, boosting token-usage forecast accuracy by **25%** with **Pandas** preprocessing
- Secured authentication with **bcryptjs**, **express-session**, **OAuth 2.0**, and **Google APIs**, while benchmarking model performance with **BLEU**, **ROUGE**, and **F1 scores** via **NLTK** and **rouge_score**

License Plate Detector Software

November 2024

- Built a **Python** video processing pipeline combining **YOLOv8 object detection** and **SORT tracking**, automating live license plate annotation and **CSV export** to streamline data processing workflows
- Trained a custom **YOLOv8** model on 22,000 images via **transfer learning**, reducing detection loss by **30%** and boosting mean average precision by **15%** over **100 epochs**
- Integrated **EasyOCR** preprocessing (grayscale, thresholding, rotation) to extract plate text with over **90%** accuracy

TMusic

May 2024

- Developed a music-social app that enabled real-time song sharing, boosting user engagement by **30%** through **PHP**, **Laravel**, **React.js**, and integration with **Spotify**, **Pusher**, and **Google OAuth APIs**
- Built back-end routes and a secure server-side database, reducing **authentication errors** by **25%** and improving data integrity using **Laravel** frameworks, **Socialite** for **OAuth**, and **Livewire** for real-time updates

LEADERSHIP EXPERIENCE

Competitive Programming Club

Toronto, ON

Toronto Metropolitan University (Previously Ryerson University)

April 2022 – May 2025

- Orchestrated weekly coding competitions, technical interview prep workshops, and peer project enhancements using **Java** and **Python**, boosting member problem-solving speed by **30%**

EDUCATION

Toronto Metropolitan University (Previously Ryerson University)

Toronto, ON

B.Sc. – Computer Engineering | Software Specialization

May 2025

- Related Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Design & Architecture, Network Security, Software Testing & QA, Intelligent Systems

SKILLS

Languages: Java, JavaScript, Python, PHP, SQL, C/C++, HTML/CSS, Assembly, VHDL, Bash

Frameworks/Technologies: React, Node.js, Laravel, Git, Jira, MySQL (relational), MongoDB, Next.js, JUnit, Express.js

Skills: API Creation, State Management, Agile & Sprint Methodologies, Prompt Engineering, Computer Vision